

NMAP SCAN REPORT

Network Scanning & Service Enumeration

Ethical Hacking Task 02 | White Band Associates

Report Information

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Organization	White Band Associates
Track	Ethical Hacking
Task	Task 02 — Network Scanning & Service Enumeration
Scan Date	2026-06-28
Scan Time	10:04 – 10:05 IST (+0530)
Target System	Kali Linux VM — localhost (127.0.0.1)
Nmap Version	7.99 (7.99+dfsg-1kali1)
Platform	x86_64-pc-linux-gnu

Part A — Nmap Installation & Version Verification

Command Used:

```
sudo apt update && sudo apt install nmap -y
nmap --version
```

Output:

```
nmap is already the newest version (7.99+dfsg-1kali1).
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1
libpcre2-10.46 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Available nsock engines: epoll poll select
```

Nmap 7.99 was already installed on the Kali Linux VM. The installation command confirmed no upgrade was needed. Version verification output confirmed the tool is compiled with full IPv6 and SSL support on a 64-bit Linux platform.

Part B — Basic Scan of Localhost

Command Used:

```
nmap localhost
```

Raw Output:

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-28 10:04 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000060s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
```

Scan Findings:

Field	Result
Host Status	Up — confirmed (latency: 0.0000060s)
Total Ports Scanned	1000 TCP ports
Open Ports	0 (None)
Port Numbers	N/A — no open ports detected
Services Running	None
Scan Duration	0.12 seconds

All 1000 TCP ports on localhost (127.0.0.1) were in a closed/reset state. This is expected for a minimal, freshly configured Kali Linux VM where no network services such as SSH, HTTP, or FTP have been started.

Part C — Service Version Detection

Command Used:

```
nmap -sV localhost
```

Raw Output:

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-28 10:05 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds
```

Service Version Table:

Port	Service	Version
—	No open ports detected	N/A

The -sV flag enables service and version detection on open ports. Since all 1000 TCP ports were closed (reset), there were no services for Nmap to interrogate. In a real-world engagement where ports are open, this scan would reveal exact service names and version numbers, which are then cross-referenced against known CVEs for vulnerability assessment.

Part D — Operating System Detection

Command Used:

```
sudo nmap -O localhost
```

Raw Output:

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-28 10:05 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000054s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 1.69 seconds
```

Question	Answer
Was OS detected?	Partially — OS detection was performed but result was inconclusive
OS Identified	Inconclusive: 'Too many fingerprints match this host to give specific OS details'
Network Distance	0 hops (loopback/localhost)
Scan Duration	1.69 seconds

Why is OS Detection Useful During Penetration Testing?

OS detection is a critical step during the enumeration phase. Knowing the target's operating system allows a tester to narrow down applicable exploits (vulnerabilities are OS-specific), identify expected default services, assess patch levels, and plan privilege escalation paths. In this case, the result was inconclusive because OS fingerprinting relies on analyzing varied TCP/IP responses from open or closed ports. With all ports closed (reset), Nmap lacked enough varied responses to distinguish between Linux kernel fingerprints with confidence. The system is confirmed to be running Kali Linux, verified from the terminal prompt and apt package manager.

Part E — Common Port Research

Port	Protocol	Purpose	Common Security Risks
20/21	FTP (TCP)	File Transfer Protocol. Port 21 = control, Port 20 = data transfer.	Plaintext credentials & data. Susceptible to brute-force, sniffing, anonymous login misconfigs.
22	SSH (TCP)	Secure Shell — encrypted remote login and command execution.	Brute-force on weak passwords. Vulnerable if outdated OpenSSH versions are used.
23	Telnet (TCP)	Remote login protocol for command-line access to a remote system.	All data including credentials sent in plaintext. Obsolete — should never be exposed.
25	SMTP (TCP)	Simple Mail Transfer Protocol — sends emails between mail servers.	Open relays can send spam/phishing. Susceptible to email spoofing and SMTP injection.
53	DNS (TCP/UDP)	Domain Name System — resolves domain names to IP addresses.	DNS spoofing, cache poisoning, amplification DDoS, zone transfer leaks via AXFR.
80	HTTP (TCP)	HyperText Transfer Protocol — serves unencrypted web pages to browsers.	Plaintext traffic vulnerable to MITM. Web apps exposed to SQLi, XSS, CSRF attacks.
110	POP3 (TCP)	Post Office Protocol v3 — downloads emails from a mail server to a client.	Credentials often in plaintext without TLS. Vulnerable to sniffing and brute-force.
143	IMAP (TCP)	Internet Message Access Protocol — manages emails directly on the server.	Similar to POP3 — plaintext without TLS. Subject to brute-force and MITM attacks.
443	HTTPS (TCP)	HTTP Secure — encrypted web communication using TLS/SSL.	SSL/TLS misconfigs, weak cipher suites, Heartbleed-style vulnerabilities, cert spoofing.
445	SMB (TCP)	Server Message Block — Windows file sharing, printer sharing, IPC.	High risk: EternalBlue (WannaCry), SMB relay attacks, credential harvesting. Never expose to internet.
3389	RDP (TCP)	Remote Desktop Protocol — graphical remote desktop access to Windows systems.	Brute-force, BlueKeep (CVE-2019-0708), credential stuffing, MITM. High-value ransomware target.

Part F — Scan Analysis

1. Which services are currently running?

Based on all three Nmap scans (basic, version detection, and OS detection), no services are currently running on this Kali Linux VM. All 1000 TCP ports on localhost (127.0.0.1) returned as closed. The host is confirmed live (latency detected), but no applications are actively listening on any port.

2. Are all open ports necessary?

Since there are no open ports on this system, there is no unnecessary exposure at this time. This represents a minimal attack surface — which is ideal from a security hardening perspective. In a typical system where services like SSH or HTTP are running, each open port should be evaluated to confirm it serves a legitimate, required purpose.

3. Which services could become security risks if misconfigured?

If services were installed and misconfigured on this system, the highest risks would come from: SSH (port 22) — brute-force target if password auth is enabled; HTTP/HTTPS (ports 80/443) — web app vulnerabilities if input validation is absent; SMB (port 445) — critical risk with weak file-sharing permissions; Telnet (port 23) — exposes all traffic in plaintext and should never be enabled.

4. Which port would you disable if it wasn't required?

Telnet (port 23) would be the first to disable. It transmits all data — including login credentials — in unencrypted plaintext, making it inherently insecure regardless of configuration. SSH (port 22) is a far superior replacement for remote access and should always be used instead.

Part G — Professional Scan Report

Commands Summary

Part	Command	Purpose	Duration
A	nmap --version	Verify Nmap installation	—
B	nmap localhost	Basic TCP port scan (top 1000 ports)	0.12s
C	nmap -sV localhost	Service and version detection	0.69s
D	sudo nmap -O localhost	Operating system fingerprinting	1.69s

Open Ports

None detected. All 1000 TCP ports on localhost (127.0.0.1) returned as closed across all scan types (basic, version detection, and OS detection).

Running Services

None detected. No listening services were identified on the target system during the scanning window.

Operating System

OS detection was performed but returned an inconclusive result: "Too many fingerprints match this host to give specific OS details" (Network Distance: 0 hops). This is a known limitation when scanning a loopback interface with no open ports — the engine cannot gather sufficient varied TCP/IP stack responses. The system is confirmed to be Kali Linux, verified from the terminal prompt and apt package output.

Observations

- The Kali Linux VM is in a minimal, hardened state with no network-facing services running.

- Scanning localhost (127.0.0.1) operates over the loopback interface, reflecting services on the machine itself — not those exposed on LAN-facing interfaces.
- Nmap OS fingerprinting requires at least one open port to generate a meaningful OS guess; with all ports closed, this feature could not return a specific result.
- All scan durations were very short (0.12s to 1.69s), consistent with scanning a local loopback interface with no open ports.

Recommendations

Priority	Recommendation
HIGH	Never enable Telnet (port 23). Use SSH for all remote access.
HIGH	If SSH is enabled, configure key-based authentication and disable password login.
MEDIUM	If a web server is started, stop it when not in use and never leave it running on a public interface.
MEDIUM	Periodically run 'nmap localhost' to audit your own system for unexpected open ports.
LOW	Remove or disable services that are installed but not actively used.

Conclusion

This task provided hands-on experience with Nmap — one of the most widely used tools in a penetration tester's toolkit. Through the scanning and enumeration phase, I learned how to identify active hosts, probe for open ports, detect running service versions, and attempt operating system fingerprinting using a structured, methodical approach.

In this exercise, the Kali Linux VM presented a clean result: no open ports and no running services. While this may initially seem like an uninteresting outcome, it actually demonstrates an important security principle — a minimal attack surface is a secure attack surface. Real-world targets are rarely this clean, and understanding what a safe baseline looks like helps in recognizing anomalies during actual engagements.

OS detection highlighted a key limitation of automated fingerprinting: tools depend on observable network behavior. When a host is quiet (no open ports), the tool cannot gather enough data to make a definitive identification. This reinforces the idea that no single tool gives a complete picture — effective reconnaissance always combines multiple techniques.

Overall, this task built a strong foundation in the scanning phase of ethical hacking and reinforced the importance of performing these activities only on authorized systems within a controlled environment.